9 (a) The nth term of a sequence is $3n + 4$

Explain why 21 is not a term of this sequence.

.....

.....

(2)

(b) Here are the first three terms of a different sequence.

1 2 4

Write down two numbers that could be the 4th term and the 5th term of this sequence.
Give the rule you have used to get your numbers.

.....

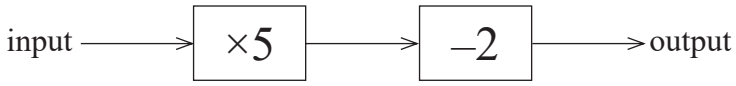
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(2)

(Total for Question 9 is 4 marks)

10 Here is a number machine.



(a) Work out the **output** when the input is 8

8 × 5 – 2 = 40 – 2 = 38

1 mark
Apply the operations in order

38

(1)

(b) Work out the **input** when the output is 28

Work backwards: 28 + 2 = 30

1 mark
Undo subtracting 2 first

30 ÷ 5 = 6

1 mark
Correct input

6

(2)

(Total for Question 10 is 3 marks)

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